Advanced Fire Fighting (AFF)

180 Practice Questions and Answers for STCW Exams

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Practice Set 1

Q1. What is the primary purpose of boundary cooling during a fire on a ship?

A) To minimize smoke production
B) To prevent fire spread by cooling adjacent surfaces
C) To increase the temperature inside the compartment
D) To accelerate the combustion process

Correct Answer: B (To prevent fire spread by cooling adjacent surfaces)

Explanation: Boundary cooling involves applying water to surfaces adjacent to a fire. Its primary purpose is to absorb heat, preventing transfer via conduction, convection, and radiation, thereby inhibiting the fire from spreading to new compartments or fuel sources. This maintains the structural integrity and confines the fire.

Q2. In advanced firefighting, what is the purpose of using a Class A foam?

A) To cool down hot surfaces
B) To suppress flammable gas release
C) To control oil and fuel fires
D) To neutralize acidic fumes

Correct Answer: C (To control oil and fuel fires)

Explanation: Class B foam is specifically designed for flammable liquid fires (oil, fuel, grease), forming a blanket that separates the fuel from oxygen and cools the surface. This creates a vapor seal to prevent reignition. Class A foam is generally for ordinary combustibles like wood or paper.

Q3. During a firefighting operation, what does the term "smoke curtain" refer to?

A) A physical barrier to prevent the entry of firefighters
B) A cloud of water droplets to control temperature
C) A curtain made of fire-resistant material to contain smoke
D) A warning signal for evacuation

Correct Answer: C (A curtain made of fire-resistant material to contain smoke)

Explanation: A smoke curtain, often part of a ship's passive fire protection system, is a flexible, fire-resistant barrier. Its purpose is to automatically deploy and contain smoke and hot gases, preventing their spread to other areas, thereby improving visibility and maintaining tenable conditions for escape.

Q4. Which firefighting equipment is specifically designed for use in confined spaces onboard a ship?

A) Fire hoses
B) Breathing apparatus
C) Foam monitors
D) Fire extinguishers

Correct Answer: B (Breathing apparatus)

Explanation: Breathing apparatus (BA) is essential for firefighters entering confined or smoke-filled spaces where oxygen levels may be depleted or toxic gases are present. It provides an independent supply of breathable air, ensuring the safety and operational effectiveness of personnel in hazardous atmospheres.

Q5. What is the primary purpose of the International Shore Connection (ISC) on a ship?

A) To supply freshwater to the crew
B) To connect with the shore power supply
C) To facilitate shore-based firefighting assistance
D) To enhance communication with nearby vessels

Correct Answer: C (To facilitate shore-based firefighting assistance)

Explanation: The International Shore Connection (ISC), mandated by SOLAS, is a standardized flange on ships. Its primary purpose is to connect the ship's fire main to a shore-based water supply or another vessel's fire main, enabling external firefighting assistance if the ship's pumps are insufficient or incapacitated.

Q6. In advanced firefighting, what is the purpose of using a water fog nozzle?

A) To create a dense fog for cooling purposes
B) To increase the range of water projection
C) To generate a fine mist for fire suppression
D) To create a visual signal for distress

Correct Answer: C (To generate a fine mist for fire suppression)

Explanation: A water fog nozzle disperses water into fine droplets, creating a large surface area for rapid heat absorption and conversion into steam. This rapidly cools the fire, displaces oxygen, and can protect firefighters from radiant heat, making it effective for general fire suppression and boundary cooling.

Q7. What is the role of a fire blanket in firefighting?

A) To smother small fires by cutting off the oxygen supply
B) To cool down hot surfaces in a fire-affected area
C) To create a barrier against radiant heat
D) To provide an escape route for trapped personnel

Correct Answer: A (To smother small fires by cutting off the oxygen supply)

Explanation: A fire blanket is used to smother small fires, particularly those involving cooking oil or clothing, by cutting off the oxygen supply. Made of fire-resistant material, it forms an airtight seal over the flames, interrupting the combustion process effectively and safely.

Q8. In the context of firefighting, what does the acronym PASS stand for?

A) Pressure Application Safety System
B) Portable Aqueous Suppression System
C) Pull, Aim, Squeeze, Sweep
D) Primary Air Supply Source

Correct Answer: C (Pull, Aim, Squeeze, Sweep)

Explanation: PASS is a mnemonic acronym for the correct sequence of operations when using a portable fire extinguisher: Pull the pin, Aim the nozzle at the base of the fire, Squeeze the handle, and Sweep from side to side. This ensures effective and safe extinguisher deployment.

Q9. What is the purpose of a fire screen in advanced firefighting?

A) To protect firefighters from radiant heat
B) To separate different compartments for ventilation
C) To extinguish fires through a fine mist
D) To detect flammable gases

Correct Answer: A (To protect firefighters from radiant heat)

Explanation: A fire screen, or thermal barrier, is utilized to shield firefighters from intense radiant heat emanating from a fire. It allows personnel to approach closer to the fire source for more effective direct attack, while significantly reducing the risk of heat-related injuries and fatigue.

Q10. When using a fire hose, what is the recommended nozzle pattern for initial cooling of a fire in a confined space?

A) Straight stream
B) Fog pattern
C) Solid stream
D) Cone spray

Correct Answer: B (Fog pattern)

Explanation: For initial cooling in a confined space, a fog pattern nozzle is recommended. The fine water droplets rapidly absorb heat and turn into steam, displacing oxygen and pushing back heat and smoke, thereby improving conditions for firefighters and preventing flashover.

Q11. In advanced firefighting, what is the primary function of a fire axe?

A) Ventilating smoke from the compartment
B) Cutting through bulkheads and doors
C) Spraying water in a concentrated stream
D) Applying foam to the fire

Correct Answer: B (Cutting through bulkheads and doors)

Explanation: A fire axe is a crucial tool for firefighters, primarily used for forcible entry, breaking through lightweight bulkheads, doors, or clearing obstructions to gain access to a fire. Its design enables effective penetration and leverage in emergency situations.

Q12. What does the term "fire tetrahedron" refer to in the context of firefighting?

A) A four-sided firefighting tool
B) The four elements required for combustion
C) A specialized firefighting drill
D) A geometric shape used in fire suppression

Correct Answer: B (The four elements required for combustion)

Explanation: The fire tetrahedron illustrates the four components necessary for fire: heat, fuel, oxygen, and a chemical chain reaction. Removing any one of these elements will extinguish the fire, which forms the fundamental principle behind all firefighting strategies and agent selection.

Q13. In the event of an electrical fire onboard, which firefighting equipment is specifically designed for use?

A) CO_2 (Carbon Dioxide) extinguisher
B) Foam extinguisher
C) Dry powder extinguisher
D) Water mist extinguisher

Correct Answer: A (CO_2 (Carbon Dioxide) extinguisher)

Explanation: CO_2 (Carbon Dioxide) extinguishers are specifically designed for electrical fires (Class C). Carbon dioxide is a non-conductive gas that smothers the fire by displacing oxygen without leaving a residue, making it ideal for protecting sensitive electrical equipment.

Q14. What is the purpose of a fireman's outfit (protective clothing) in advanced firefighting?

A) To identify firefighting personnel
B) To enhance physical agility
C) To provide protection from heat and flame
D) To store firefighting tools

Correct Answer: C (To provide protection from heat and flame)

Explanation: A fireman's outfit, mandated by SOLAS, provides essential personal protective equipment (PPE) for firefighters. It is constructed from heat-resistant materials to shield the wearer from high temperatures, flames, radiant heat, and minor mechanical hazards, ensuring safety during firefighting operations.

Q15. In the context of fire safety, what does the acronym "RECAAR" stand for?

A) Rapid Escape and Control Against Radiant heat
B) Rescue, Evacuation, and Containment of Aerial Risks
C) Response, Escape, and Control Against Adverse Reactions
D) Rescue, Evacuation, Containment, Assessment, and Recovery

Correct Answer: D (Rescue, Evacuation, Containment, Assessment, and Recovery)

Explanation: RECAAR is a structured approach to emergency response and incident management, particularly in maritime contexts. It stands for Rescue, Evacuation, Containment (of the incident), Assessment (of damage/situation), and Recovery (returning to normalcy), guiding comprehensive emergency procedures.

Q16. What is the purpose of a flame arrester in firefighting?

A) To control the spread of flammable liquids
B) To arrest the progression of flames in a compartment
C) To provide visual cues during firefighting operations
D) To enhance communication among firefighters

Correct Answer: A (To control the spread of flammable liquids)

Explanation: A flame arrester is a device fitted to tank vents or pipes carrying flammable vapors. Its primary purpose is to prevent the propagation of a flame through a flammable gas mixture by rapidly cooling the flame below its ignition temperature, thus stopping flashback or explosion.

Q17. During a shipboard fire emergency, what is the primary purpose of the muster list?

A) To account for all personnel and passengers
B) To provide instructions for firefighting techniques
C) To allocate firefighting responsibilities
D) To identify the location of fire hydrants

Correct Answer: A (To account for all personnel and passengers)

Explanation: The muster list, as per SOLAS, is a crucial document detailing duties for crew members in various emergencies, including fire. Its primary purpose during an emergency muster is to ensure all personnel and passengers are accounted for and to verify their safety and assigned roles.

Q18. What is the recommended method for handling an injured person during a shipboard firefighting operation?

A) Immediate evacuation to the shore
B) Placing the person in a confined space
C) Moving the person to a designated safe area
D) Administering first aid at the fire location

Correct Answer: C (Moving the person to a designated safe area)

Explanation: During a shipboard fire, the immediate priority for an injured person is to move them quickly and safely to a designated safe area, away from direct fire hazards or smoke. Administering first aid should follow once the individual is in a secure environment.

Q19. Which firefighting agent is commonly used to suppress fires involving flammable metals?

A) Water
B) Foam
C) Dry powder
D) CO_2 (Carbon Dioxide)

Correct Answer: C (Dry powder)

Explanation: Dry powder extinguishers (Class D) are specifically designed for fires involving combustible metals like magnesium, titanium, or sodium. These specialized powders smother the fire and form a crust that excludes oxygen, effectively suppressing these unique and intense fires.

Q20. What is the purpose of a fire control plan on a ship?

A) To outline emergency escape routes for passengers
B) To specify the locations of fire alarm pull stations
C) To provide guidance for controlling and suppressing fires
D) To identify the ship's navigation routes

Correct Answer: C (To provide guidance for controlling and suppressing fires)

Explanation: The fire control plan, prominently displayed on ships as per SOLAS, provides critical information on the ship's fire protection arrangements. It guides effective fire control and suppression by indicating locations of fire-extinguishing appliances, fire-resistant divisions, and means of escape.

Q21. What is the purpose of a deluge system in shipboard firefighting?

A) To deliver a high-pressure water mist
B) To flood a compartment with water rapidly
C) To control electrical fires with a dry powder
D) To generate foam for flammable liquid fires

Correct Answer: B (To flood a compartment with water rapidly)

Explanation: A deluge system is a fixed fire suppression system designed to rapidly discharge a large volume of water over a protected area or into a compartment. It's used in high-hazard areas to quickly cool and smother large fires or create an effective water barrier.

Q22. During a shipboard firefighting operation, what is the primary function of a portable gas detector?

A) To locate fire hydrants
B) To identify flammable gases
C) To measure ambient temperature
D) To monitor radio communication

Correct Answer: B (To identify flammable gases)

Explanation: A portable gas detector is a vital piece of equipment used to monitor the atmosphere for the presence of flammable, toxic, or oxygen-deficient gases. It ensures the safety of personnel entering confined spaces or areas potentially affected by gas leaks during a fire emergency.

Q23. In the context of advanced firefighting, what is the purpose of a water curtain?

A) To create a barrier against radiant heat
B) To supply water to firefighting hoses
C) To generate foam for liquid fires
D) To create a visual signal for distress

Correct Answer: A (To create a barrier against radiant heat)

Explanation: A water curtain, formed by a series of nozzles spraying water, creates a dense sheet of water. This acts as a thermal barrier, absorbing radiant heat and protecting adjacent areas or personnel from intense heat radiation, thereby preventing fire spread or providing a safe egress path.

Q24. Which firefighting equipment is commonly used to cut ventilation openings in ship compartments?

A) Fire axe
B) Water fog nozzle
C) Thermal imaging camera
D) Hydraulic rescue tool

Correct Answer: D (Hydraulic rescue tool)

Explanation: Hydraulic rescue tools, such as spreaders and cutters, are powerful implements used for heavy-duty cutting and forcing open structural components. They are highly effective for cutting ventilation openings in robust ship compartments or structures where conventional tools are insufficient.

Q25. During a shipboard firefighting operation, what is the primary purpose of a positive pressure breathing apparatus (PPBA)?

A) To supply fresh air to the compartment
B) To prevent smoke from entering the mask
C) To maintain positive pressure inside the mask
D) To enhance communication among firefighters

Correct Answer: C (To maintain positive pressure inside the mask)

Explanation: A positive pressure breathing apparatus (PPBA) maintains a slightly higher pressure inside the mask than the ambient pressure. This crucial feature prevents toxic smoke or gases from leaking into the mask, even with minor seal imperfections, ensuring the wearer breathes only clean, supplied air.

Q30. In advanced firefighting, what is the primary purpose of a fire-resistant bulkhead?

A) To provide a visual barrier against smoke
B) To contain the spread of fire within compartments
C) To enhance communication among firefighting teams
D) To store firefighting equipment safely

Correct Answer: B (To contain the spread of fire within compartments)

Explanation: Fire-resistant bulkheads are structural divisions designed to withstand fire for a specified period (e.g., A-60, B-15). Their primary purpose, as per SOLAS, is to contain a fire within its compartment of origin, preventing its spread to other parts of the ship and protecting escape routes.

Practice Set 2

Q1. A safety is provided on the CO2 discharge piping to prevent:

A) Flooding of a space where personnel are present
B) Over pressurization of the CO2 discharge piping
C) Rupture of cylinder due to temperature increase
D) Over pressurization of the space being flooded

Correct Answer: B (Over pressurization of the CO2 discharge piping)

Explanation: A safety relief device, often a bursting disc, is installed on CO2 discharge piping to prevent over-pressurization. In the event of an abnormal pressure build-up, it ruptures to release the CO2, protecting the piping system from damage or catastrophic failure, compliant with the FSS Code.

Q2. Apart from red what other color is frequently used for a CO2 extinguisher:

A) Blue
B) Yellow
C) Black
D) Red

Correct Answer: C (Black)

Explanation: While the main body of fire extinguishers is typically red, different types are identified by a color band or the entire body color. For CO2 extinguishers, the body is often black or has a black band, clearly distinguishing them from other types for rapid identification.

Q3. Flue gases are used directly for inerting?

A) True
B) False

Correct Answer: B (False)

Explanation: False. Flue gases are not used directly for inerting. They must first be processed by an inert gas generator system, which cleans, cools, and removes sulfur oxides and particulate matter, before being supplied to cargo tanks for inerting purposes to prevent explosions.

Q4. In fighting a fire in a fuel tank, the first action you should attempt is to:

A) Station someone at the fixed CO2 release controls
B) Top off the tank to force out all vapors
C) Begin transferring the fuel to other tanks
D) Secure all sources of fresh air to the tank

Correct Answer: D (Secure all sources of fresh air to the tank)

Explanation: For a fire in a fuel tank, the first critical step is to eliminate the oxygen supply by securing all openings, vents, and fresh air sources. This starves the fire, helping to contain it before applying extinguishing agents and mitigating the risk of rapid spread.

Q5. The FIRST course of action in fighting a fire in a cargo or fuel oil tank is to:

A) Spray the tank boundaries with a fire hose to promote cooling
B) Direct a fire hose into the tank and energize the fire main
C) Discharge an initial charge of CO2 with a hand portable extinguisher
D) Secure all openings to the tank

Correct Answer: D (Secure all openings to the tank)

Explanation: Securing all openings to a cargo or fuel oil tank immediately cuts off the oxygen supply, a critical element for combustion. This primary action helps to contain the fire, reduces its intensity, and prepares the space for the safe and effective application of fixed extinguishing systems.

Q6. Three side of fire triangle are fuel, heat and ....?

A) Petrol
B) Gases
C) Oxygen
D) Combustible material

Correct Answer: C (Oxygen)

Explanation: The fire triangle comprises three essential elements for combustion: fuel (combustible material), heat (ignition source), and oxygen (oxidizer). Removing or disrupting any one of these elements will effectively extinguish the fire, a fundamental concept in firefighting theory and practice.

Q7. Why should not carbon dioxide extinguisher be used in accommodation space:

A) Frostbite risk
B) It can displace the oxygen, causing Asphyxiation
C) Health hazard

Correct Answer: B (It can displace the oxygen, causing Asphyxiation)

Explanation: CO2 displaces oxygen to smother fires. However, in confined spaces like accommodation areas, this oxygen displacement poses a severe risk of asphyxiation to personnel. Therefore, CO2 fixed systems are generally restricted to machinery spaces or cargo holds after personnel evacuation.

Q8. In a fixed CO2 fire extinguishing system where pressure from pilot cylinders is used to release the CO2 from the main bank of cylinders, the number of required pilot cylinders shall be at least:

A) 6
B) 4
C) 3
D) 2

Correct Answer: D (2)

Explanation: According to SOLAS regulations and the FSS Code, in a fixed CO2 system where pilot cylinders release CO2 from the *main bank, there must be at least two pilot cylinders. This redundancy ensures system reliability, so a single pilot cylinder failure does not render the entire system inoperable.*

Q9. Fixed CO2 systems would not be used on crew's quarters or:

A) the engine room
B) cargo holds
C) spaces open to the atmosphere
D) the paint locker

Correct Answer: C (spaces open to the atmosphere)

Explanation: Fixed CO2 systems work by displacing oxygen to smother a fire. They are ineffective and potentially dangerous in spaces open to the atmosphere, such as open decks or other unenclosed areas, as the CO2 would rapidly dissipate, failing to achieve the necessary concentration to extinguish the fire.

Q10. Your vessel is equipped with a fixed CO2 system and a fire main system. In the event of an electrical fire in the engine room what is the correct procedure for fighting the fire?

A) Evacuate the engine room and use the fire main system.
B) Evacuate the engine room and use the CO2 system.
C) Use the fire main system and evacuate the engine room.
D) Use the CO2 system and evacuate the engine room.

Correct Answer: B (Evacuate the engine room and use the CO2 system.)

Explanation: For an electrical fire in the engine room, personnel must first evacuate due to the danger of toxic smoke and CO2 deployment. Subsequently, the fixed CO2 system should be released, as CO2 is non-conductive and effectively smothers electrical fires without causing further damage to equipment.

Q11. When pilot cylinder pressure is used as a means to release the CO2 from a fixed fire extinguishing system consisting of four storage cylinders, the number of pilot cylinders shall be at least:

A) 1
B) 4
C) 3
D) 2

Correct Answer: D (2)

Explanation: As per SOLAS and FSS Code requirements for fixed CO2 systems, if pilot cylinders are used to release the CO2 from the main storage cylinders, a minimum of two pilot cylinders are mandatory. This ensures critical operational redundancy and system reliability, preventing a single point of failure.

Q12. After extinguishing a paint locker fire using the fixed CO2 system, the next action is to have the space:

A) doused with water to prevent reflash
B) checked for oxygen content
C) left closed with vents off until all boundaries are cool
D) opened and burned material removed

Correct Answer: C (left closed with vents off until all boundaries are cool)

Explanation: After a CO2 discharge, the space should remain closed with ventilation secured until all boundaries have cooled down. Opening the space too soon can introduce oxygen, potentially causing re-ignition or flashover due to residual heat, a critical post-fire procedure to prevent secondary incidents.

Q13. In a fixed CO_2 extinguishing system where provision is made for the release of CO_2 by operation of a remote control, provision shall also be made for releasing the CO_2:

A) at the cylinder location
B) from the cargo control station
C) from the bridge
D) from inside the engine room

Correct Answer: A (at the cylinder location)

Explanation: SOLAS regulations require that fixed CO2 systems with remote release mechanisms also have a provision for local release at the cylinder location. This ensures the system can still be activated manually in case of a remote control malfunction, providing a critical backup and enhancing system reliability.

Q14. Which of the following statements is FALSE, concerning the regulations pertaining to the cylinder room of a fixed CO_2 fire extinguishing system?

A) The temperature of the room should never exceed 130 degreesF.
B) The door must be kept unlocked.
C) The compartment must be properly ventilated.
D) The compartment shall be clearly marked and identifiable.

Correct Answer: B (The door must be kept unlocked.)

Explanation: The statement that the door must be kept unlocked is FALSE. The CO2 cylinder room must be kept locked to prevent unauthorized access, but with clear instructions and readily available keys for authorized personnel. SOLAS and FSS Code mandate proper ventilation, clear marking, and temperature control.

Q15. You are releasing carbon dioxide gas (CO_2) into an engine compartment to extinguish a fire. The CO_2 will be most effective if the:

A) air flow to the compartment is increased with blowers
B) compartment is closed and airtight
C) compartment is left open to the air
D) compartment is closed and ventilators are opened

Correct Answer: B (compartment is closed and airtight)

Explanation: To extinguish a fire with CO2, the compartment must be sealed to contain the gas, thereby reducing the oxygen concentration to a level below that required for combustion. Increasing air flow or leaving the compartment open would disperse the CO2, rendering it ineffective.

Q16. CO_2 cylinders, which protect the small space in which they are stored must:

A) NOT contain more than 200 pounds of CO_2
B) have an audible alarm
C) be automatically operated by a heat actuator
D) All of the above

Correct Answer: A (NOT contain more than 200 pounds of CO_2)

Explanation: SOLAS regulations require CO2 cylinders for cargo spaces to not exceed 200 pounds of CO2 for safety and regulatory compliance. While alarms are important, this specific weight limit is a key requirement for these cylinders in such applications.

Q17. In areas where CO_2 piping is installed, such piping may not be used for any other purpose EXCEPT:

A) to run the emergency wiring to the space
B) to ventilate the space
C) in connection with the water sprinkler system
D) in connection with the fire-detecting system

Correct Answer: D (in connection with the fire-detecting system)

Explanation: Piping for fixed CO_2 fire extinguishing systems is dedicated solely to its fire suppression function to ensure reliability and prevent contamination or obstruction. While fire detection systems share an area, CO_2 piping cannot be used for ventilation or other unrelated purposes.

Q18. Control valves of a CO_2 system may be located within the protected space when:

A) an automatic heat-sensing trip is installed
B) the CO_2 cylinders are also in the space
C) there is also a control valve outside
D) it is impractical to locate them outside

Correct Answer: C (there is also a control valve outside)

Explanation: Control valves for CO_2 systems in protected spaces are typically located outside. However, if impractical, they may be located inside provided a secondary control valve is also present externally for safe manual operation during an emergency.

Q19. While you are working in a space, the fixed CO_2 system is accidentally activated. You should:

A) make sure all doors and vents are secured
B) retreat to fresh air and ventilate the compartment before returning
C) continue with your work as there is nothing you can do to stop the flow of CO_2
D) secure the applicators to preserve the charge in the cylinders

Correct Answer: B (retreat to fresh air and ventilate the compartment before returning)

Explanation: If a CO_2 system activates accidentally while you are in the space, the immediate priority is personal safety. Retreating to fresh air and ensuring the compartment is ventilated before re-entry is crucial as CO_2 displaces oxygen, posing an asphyxiation risk.

Q20. Large volumes of carbon dioxide are safe and effective for fighting fires in enclosed spaces, such as in a pumproom, provided that the:

A) ventilation system is kept operating
B) ventilation system is secured and all persons leave the space
C) persons in the space wear damp cloths over their mouths and nostrils
D) persons in the space wear gas masks

Correct Answer: B (ventilation system is secured and all persons leave the space)

Explanation: For effective CO_2 fire suppression in enclosed spaces like pumprooms, the ventilation system must be secured to maintain the required concentration of CO_2. This also necessitates ensuring all personnel have evacuated the space prior to discharge.

Q21. Automatic mechanical ventilation shutdown is required for CO_2 systems protecting the:

A) galley
B) living quarters
C) cargo compartments
D) machinery spaces

Correct Answer: D (machinery spaces)

Explanation: Automatic mechanical ventilation shutdown is mandatory for CO_2 systems protecting machinery spaces to ensure the inerting effect of the CO_2 is maintained. This prevents dilution and maximizes the effectiveness of the extinguishing agent.

Q22. Carbon dioxide as a fire fighting agent has which advantage over other agents?

A) It causes minimal damage.
B) It is safer for personnel.
C) It is cheaper.
D) It is most effective on a per unit basis.

Correct Answer: A (It causes minimal damage.)

Explanation: Carbon dioxide offers the advantage of being a clean agent, meaning it leaves no residue after discharge. This is particularly beneficial for protecting sensitive equipment where water or foam might cause collateral damage.

Q23. CO_2 cylinders forming part of a fixed fire extinguishing system must be pressure tested at least every:

A) 12 years
B) 6 years
C) 2 years
D) year

Correct Answer: A (12 years)

Explanation: SOLAS regulations mandate that CO_2 cylinders used in fixed fire extinguishing systems must undergo hydrostatic testing every 12 years to ensure their structural integrity and safety for continued use.

Q24. Fixed carbon dioxide extinguishing systems, for machinery spaces that are normally manned, are actuated by one control to open the stop valve in the line leading to the space, and:

A) three separate controls to release the CO_2
B) two separate controls to release the CO_2
C) a separate control to release the CO_2
D) the same control releasing the CO_2

Correct Answer: B (two separate controls to release the CO_2)

Explanation: For normally manned machinery spaces protected by a CO_2 system with a stop valve, a two-control operation is required. One control initiates the process (e.g., opening the stop valve), and a second control actually releases the CO_2.

Q25. A fixed carbon dioxide extinguishing system for a machinery space, designed WITHOUT a stop valve in the line leading to the protected space, is actuated by:

A) one control
B) two controls
C) three controls
D) none of the above

Correct Answer: A (one control)

Explanation: If a CO_2 system for a machinery space is designed without a stop valve in the line to the protected space, the release of CO_2 is achieved through a single control mechanism. This simplifies the activation process in such configurations.

Q26. On cargo vessels, the discharge of the required quantity of carbon dioxide into any 'tight' space shall be completed within:

A) 1 minute
B) 2 minutes
C) 4 minutes
D) 6 minutes

Correct Answer: B (2 minutes)

Explanation: SOLAS requires that the discharge of the designed quantity of carbon dioxide into a 'tight' space on cargo vessels be completed within 2 minutes. This rapid discharge is essential for quickly suppressing the fire before it escalates.

Q27. Aboard a cargo vessel, the supply of carbon dioxide used in a fixed extinguishing system MUST at least be sufficient for what space:

A) the space requiring the largest amount
B) the engine room and largest cargo space
C) all cargo spaces
D) all the spaces of a vessel

Correct Answer: A (the space requiring the largest amount)

Explanation: The supply of CO2 for a fixed extinguishing system on a cargo vessel must be sufficient to protect the single space that requires the largest quantity of gas. This ensures adequate protection for the most demanding scenario.

Q28. How often must CO2 systems be inspected to confirm cylinders are within 10% of the stamped full weight of the charge:

A) quarterly
B) semiannually
C) annually
D) biannually

Correct Answer: C (annually)

Explanation: CO2 cylinders in fixed fire extinguishing systems are required to be inspected annually. This inspection includes weighing the cylinders to confirm they are within 10% of the stamped full weight of the charge.

Q29. In weighing CO2 cylinders, they must be recharged if weight loss exceeds:

A) 10% of weight of charge
B) 20% of weight of charge
C) 15% of weight of full bottle
D) 10% of weight of full bottle

Correct Answer: A (10% of weight of charge)

Explanation: CO2 cylinders for fixed fire extinguishing systems must be recharged if their weight loss exceeds 10% of the original weight of the charge. This ensures the system contains the correct amount of agent for effective fire suppression.

Q30. Before using a fixed CO2 system to fight an engine room fire, you must:

A) secure the engine room ventilation
B) secure the machinery in the engine room
C) evacuate all engine room personnel
D) All of the above

Correct Answer: D (All of the above)

Explanation: Before activating a fixed CO2 system in an engine room, essential safety precautions include securing the ventilation to contain the CO2, stopping the machinery to prevent further hazards, and evacuating all personnel to prevent asphyxiation.

Q31. The CO2 flooding system is actuated by a sequence of steps which are:

A) open stop valve, open control valve, trip alarm
B) open bypass valve, break glass, pull handle
C) sound evacuation alarm, pull handle
D) break glass, pull valve, break glass, pull cylinder control

Correct Answer: D (break glass, pull valve, break glass, pull cylinder control)

Explanation: The typical sequence for actuating a fixed CO2 flooding system involves a series of steps to ensure safety and proper function: breaking the glass to access controls, pulling a handle or lever to release the CO2, and confirming cylinder control activation.

Q32. Which advantage does dry chemical have over carbon dioxide (CO2) in firefighting?

A) Compatible with all foam agents
B) More protective against re-flash
C) Cleaner
D) All of the above

Correct Answer: B (More protective against re-flash)

Explanation: Dry chemical agents offer a greater advantage over CO2 in preventing re-flash because they form a crust on the burning material, effectively isolating it from oxygen. CO2 primarily works by diluting oxygen and cooling.

Q33. What is NOT a characteristic of carbon dioxide fire-extinguishing agents?

A) Effective even if ventilation is not shut down
B) Will not deteriorate in storage
C) Non-corrosive
D) Effective on electrical equipment

Correct Answer: A (Effective even if ventilation is not shut down)

Explanation: Carbon dioxide is not effective if ventilation is not shut down, as its extinguishing action relies on displacing oxygen and cooling. The other listed characteristics-stability, non-corrosiveness, and effectiveness on electrical fires-are indeed advantages of CO2.

Q34. In a fixed carbon dioxide extinguishing system for a machinery space, designed WITH a stop valve in the line leading to the protected space, the flow of CO2 is established by actuating:

A) three controls
B) two controls
C) one control
D) none of the above

Correct Answer: B (two controls)

Explanation: When a fixed CO2 system for a machinery space includes a stop valve in the line to the protected space, its activation requires two controls. One control operates the stop valve, and the second control releases the CO2 from the cylinders.

Practice Set 3

Q1. An important step in fighting any electrical fire is to:

A) Apply water fog immediately
B) De-energize the circuit
C) Increase ventilation
D) Use chemical foam

Correct Answer: B (De-energize the circuit)

Explanation: De-energizing the circuit is the most critical step when fighting an electrical (Class C) fire. This removes the power source, preventing electric shock hazards and the risk of reignition.

Q2. Which fire detection system is actuated by sensing a heat rise in a compartment?

A) Flame detection system
B) Manual pull stations
C) Automatic fire detection system
D) Smoke obstruction system

Correct Answer: C (Automatic fire detection system)

Explanation: An automatic fire detection system can be actuated by sensing a heat rise, which triggers an alarm or the activation of a fire suppression system. Other systems listed detect flame, require manual activation, or sense smoke particles.

Q3. Water fog is an effective firefighting agent because it:

A) Has a great cooling ability
B) Breaks the molecular chain reaction chemically
C) Displaces oxygen completely
D) Smothers the fire using dense foam

Correct Answer: A (Has a great cooling ability)

Explanation: Water fog is effective because its fine spray creates a large surface area, maximizing heat absorption and thus providing significant cooling. This cooling effect is crucial in suppressing fires of Class A and B types.

Q4. A fire of escaping liquefied flammable gas is best extinguished by:

A) Flooding with low-velocity fog
B) Stopping the flow of gas
C) Applying large amounts of dry chemical powder
D) Cooling the surrounding storage cylinders

Correct Answer: B (Stopping the flow of gas)

Explanation: The best way to extinguish a fire involving escaping liquefied flammable gas is to stop the flow of the gas itself. Applying extinguishing agents can be dangerous and may not effectively stop the fuel source.

Q5. When combating a Class C fire, which of the following dangers may be present?

A) Spontaneous combustion of scrap metal
B) High expansion foam flashover
C) Toxic fumes from burning insulation or electric shock
D) Exploding fuel-air vapor mixtures

Correct Answer: C (Toxic fumes from burning insulation or electric shock)

Explanation: Class C fires, involving energized electrical equipment, pose the risk of toxic fumes from burning insulation materials and the immediate danger of electric shock to personnel.

Q6. The lowest temperature at which the vapors of a flammable liquid will ignite and cause self-sustained combustion in the presence of a spark or flame is the:

A) Flash Point
B) Ignition Point
C) Fire Point
D) Critical Point

Correct Answer: C (Fire Point)

Explanation: The Fire Point is the lowest temperature at which a liquid will give off vapors that can sustain combustion after ignition. This is distinct from the Flash Point, which is the temperature at which vapors ignite momentarily.

Q7. The fire protection provided for the propulsion motor and generator of a diesel-electric drive vessel is usually a:

A) Fixed foam system
B) Fixed CO_2 system
C) Water sprinkler system
D) Dry powder station

Correct Answer: B (Fixed CO_2 system)

Explanation: Diesel-electric drive vessels commonly use fixed CO_2 systems to protect propulsion motors and generators. CO_2 is suitable for electrical equipment as it is non-conductive and leaves no residue.

Q8. Metal fires should be fought using which extinguisher type?

A) Carbon dioxide
B) Aqueous film-forming foam (AFFF)
C) Dry powder
D) Halon gas

Correct Answer: C (Dry powder)

Explanation: Metal fires (Class D) require specialized extinguishing agents like dry powder. Agents such as CO_2, AFFF, or Halon can react violently with burning metals, intensifying the fire or causing explosions.

Q9. Which type of portable fire extinguisher is best suited for putting out a Class D fire?

A) Carbon dioxide
B) Water spray
C) Dry powder
D) Dry chemical

Correct Answer: C (Dry powder)

Explanation: Class D fires, involving combustible metals, require specific dry powder extinguishers designed to smother and cool the metal. CO_2 and dry chemical are generally unsuitable and can be hazardous for metal fires.

Q10. Class 1 watertight doors are:

A) Hinged type
B) Vertical sliding type
C) Horizontal sliding type
D) Rolling type

Correct Answer: A (Hinged type)

Explanation: Class 1 watertight doors are hinged types, typically found in passenger ships, designed to be quickly closed. Class 2 doors are usually sliding types and are often operated remotely.

Q11. A fire in a fuel oil settling tank is a:

A) Class A fire
B) Class B fire
C) Class C fire
D) Class D fire

Correct Answer: B (Class B fire)

Explanation: A fire in a fuel oil settling tank involves flammable liquids, classifying it as a Class B fire. Class A involves ordinary combustibles, Class C electrical, and Class D metals.

Q12. The symbols for Fire Control plans are approved by which organization?

A) International Labor Organization (ILO)
B) International Maritime Organization (IMO)
C) National Fire Protection Association (NFPA)
D) International Standards Organization (ISO)

Correct Answer: B (International Maritime Organization (IMO))

Explanation: The International Maritime Organization (IMO) is responsible for approving the symbols used on Fire Control plans, ensuring standardized representation of fire safety equipment and arrangements across vessels.

Q13. Which dangerous condition should be kept in mind after using a 6.8 kg CO2 fire extinguisher on a small oil fire on the engine room floor plates?

A) Suffocation inside the main casing
B) Rupture of downstream valves
C) Rekindling of the fire
D) Freezing of mechanical controls

Correct Answer: C (Rekindling of the fire)

Explanation: After using a CO2 extinguisher on an oil fire, rekindling is a potential danger as the CO2 dissipates and the hot oil may reach its ignition temperature again if not properly cooled or smothered.

Q14. What emergency equipment is NOT found in the crew mess in the view of a Fire Control plan?

A) Portable fire extinguisher
B) Heat sensor
C) Fire axe
D) Emergency escape breathing device (EEBD)

Correct Answer: B (Heat sensor)

Explanation: Heat sensors are part of an automatic fire detection system and are typically located in the protected spaces themselves, not in general areas like the crew mess. Other listed items are emergency equipment often found in such locations.

Q15. Dry powder fire extinguishers which contain a mixture of graphite and sodium chloride as firefighting agents are generally used to fight which type of fire?

A) Class A wood fire
B) Class B fuel oil fire
C) Class C electrical fire
D) Class D metal fire

Correct Answer: D (Class D metal fire)

Explanation: Dry powder extinguishers containing graphite and sodium chloride are specifically designed for Class D fires, which involve combustible metals. These agents form a crust, smothering the fire and preventing re-ignition.

Q16. On international voyages, tank ships of 500 gross tons or more are required to have facilities to enable connection on each side of the ship for this fire control equipment:

A) International Shore connection
B) High expansion foam generator
C) Fixed sprinkler bypass valve
D) Emergency bilge pump suction manifold

Correct Answer: A (International Shore connection)

Explanation: On oil tankers of 500 gross tons or more, an International Shore Connection is required on each side of the ship. This allows shore-based fire fighting equipment to connect to the vessel's fire main system.

Q17. Mechanical foam used for firefighting is produced by mechanically mixing and agitating:

A) Foam chemical with nitrogen gas
B) Liquid CO2 with water solution
C) Foam chemical with air and water
D) Dry chemical powder with water stream

Correct Answer: C (Foam chemical with air and water)

Explanation: Mechanical foam used in firefighting is generated by mixing a foam concentrate with water and then agitating this mixture with air. This process creates a foam blanket that cools and smothers the fire.

Q18. What must be mounted at a small passenger vessel's operating station for use by the master and crew?

A) Vessel maintenance record
B) Emergency instructions
C) SOLAS training booklet
D) International code of signals chart

Correct Answer: B (Emergency instructions)

Explanation: Emergency instructions detailing actions to be taken during emergencies must be prominently displayed at the vessel's operating station for immediate access by the master and crew.

Q19. The fire main system should be flushed with fresh water whenever possible to:

A) Minimize downstream supply pressure
B) Help destroy marine growth
C) Lubricate fire pump impellers
D) Reduce systemic scale formation

Correct Answer: B (Help destroy marine growth)

Explanation: Flushing the fire main system with fresh water after use helps to reduce systemic scale formation by removing corrosive saltwater. This prolongs the life and efficiency of the piping and associated equipment.

Q20. The muster list shows each rig hand's muster station, his duties during abandonment, basic instructions, and:

A) All emergency signals
B) Navigation course parameters
C) Next port of call logistics
D) Engine room operating instructions

Correct Answer: A (All emergency signals)

Explanation: A muster list, as required by SOLAS, details not only muster stations and duties but also all relevant emergency signals to ensure all crew are aware of the different alarm sounds and their meanings.

Q21. If a fire ignites in the engine room as a result of a high-pressure fuel oil leak, you should first:

A) Evacuate all personnel immediately
B) Release the fixed CO2 system
C) Shut off the fuel oil supply
D) Direct a stream of water at the flames

Correct Answer: C (Shut off the fuel oil supply)

Explanation: The immediate priority in a fuel oil leak fire is to stop the source of fuel. Shutting off the fuel supply (Option C) removes the fuel element of the fire triangle, thus allowing the fire to extinguish. Releasing CO2 (Option B) or using water (Option D) are secondary actions, and evacuation (Option A) is appropriate if the fire cannot be controlled.

Q22. If there has been a fire in a closed, unventilated compartment, it may be unsafe to enter because of:

A) High accumulations of nitrogen gas
B) Excess relative humidity
C) A lack of oxygen
D) Built-up electrostatic charges

Correct Answer: C (A lack of oxygen)

Explanation: Fires consume oxygen for combustion. In a closed compartment, the oxygen level will significantly decrease, creating an atmosphere with insufficient oxygen to support human life. Therefore, entering such a compartment without self-contained breathing apparatus (SCBA) is extremely hazardous due to potential asphyxiation.

Q23. Which of the listed classes of fires can best be extinguished with a water fog?

A) Class A and B
B) Class B and C
C) Class C and D
D) Class A and D

Correct Answer: A (Class A and B)

Explanation: Water fog is particularly effective for Class A fires (ordinary combustibles like wood, paper, cloth) by cooling and penetrating the material, and for Class B fires (flammable liquids) by cooling the liquid surface to below its flashpoint and forming a barrier. It is not suitable for Class C (electrical) or Class D (combustible metals) fires.

Q24. The fixed CO2 fire extinguishing system has been activated to extinguish a large engine room bilge fire. When is the best time to vent the combustible products from the engine room?

A) Immediately after the fire goes out
B) After the metal surfaces have cooled down
C) Right before entering the space with an SCBA
D) One hour after initial discharge regardless of temperature

Correct Answer: B (After the metal surfaces have cooled down)

Explanation: After a CO2 discharge, hot surfaces can re-ignite vapours. Venting the compartment after the metal surfaces have cooled down (Option B) reduces the risk of re-ignition and allows hazardous gases to dissipate safely before entry, ideally while wearing SCBA.

Q25. A definite advantage of using water as a firefighting agent is its characteristic of:

A) Breaking down chemical bonding chains safely
B) Non-conductivity near live machinery
C) Rapid expansion as water absorbs heat and changes to steam
D) Leaving clean residue that acts as a fire barrier

Correct Answer: C (Rapid expansion as water absorbs heat and changes to steam)

Explanation: Water's high latent heat of vaporization means it absorbs a significant amount of heat as it turns to steam. This rapid expansion (Option C) helps to cool the burning material and can also help to push flames away from the surface.

Q26. What is NOT a characteristic of a carbon dioxide fire extinguishing agent?

A) It leaves no messy residue after use
B) It is effective even if ventilation is not shut down
C) It displaces air to smother a fire
D) It can cause frostbite upon contact with skin

Correct Answer: B (It is effective even if ventilation is not shut down)

Explanation: Carbon dioxide (CO2) extinguishers displace oxygen to smother a fire, and as they discharge, the rapid expansion of gas causes extreme cooling, potentially leading to frostbite. However, CO2 is electrically conductive and must not be used on live electrical equipment if safer alternatives exist, and ventilation is crucial after use to clear the oxygen-deficient atmosphere.

Q27. It is necessary to secure the forced ventilation to a compartment where there is a fire to:

A) Prevent additional oxygen from reaching the fire
B) Retain toxic gases to help smother flames
C) Protect the ventilation fan motor from overspeeding
D) Limit the expansion of cooling steam lines

Correct Answer: A (Prevent additional oxygen from reaching the fire)

Explanation: Forcing ventilation into a compartment with a fire provides an additional supply of oxygen, which fuels the combustion process. Securing the ventilation (Option A) removes this oxygen source, aiding in fire suppression, consistent with removing an element of the fire triangle.

Q28. The four basic components of a fire are a chain reaction, heat, fuel, and:

A) Nitrogen
B) Carbon dioxide
C) Oxygen
D) Carbon monoxide

Correct Answer: C (Oxygen)

Explanation: The fire triangle, representing the essential components for combustion, consists of fuel, heat, and oxygen. The chain reaction is often considered the fourth element in the fire tetrahedron, but oxygen (Option C) is the fundamental component that sustains the combustion process.

Q29. The flammable limits of an atmosphere are the:

A) Maximum temperatures at which an oil source produces vapors
B) Upper and lower percentage of vapor concentrations in an atmosphere which will burn if an ignition source is present
C) Safe ambient operating thresholds for storage facilities
D) Minimum moisture concentrations needed to halt spontaneous ignition

Correct Answer: B (Upper and lower percentage of vapor concentrations in an atmosphere which will burn if an igniti

Explanation: Flammable limits define the range of concentrations of a flammable gas or vapor in air that will burn or explode when an ignition source is present. The lower flammable limit (LFL) is the minimum concentration, and the upper flammable limit (UFL) is the maximum concentration.

Q30. A fire must be ventilated:

A) To supply additional oxygen to speed up burning
B) To prevent the gases of combustion from surrounding the firefighters
C) Immediately after triggering a fixed gas smothering system
D) To cool down fuel reservoirs below their flashpoints

Correct Answer: B (To prevent the gases of combustion from surrounding the firefighters)

Explanation: Ventilation is crucial in firefighting to remove smoke and toxic gases produced by combustion, ensuring a safer environment for firefighters to operate and preventing the spread of toxic fumes to other areas of the vessel. Supplying extra oxygen (Option A) would worsen the fire.

Practice Set 4

Q1. A fire in a radio transmitter would be of what class under NFPA?

A) Class A
B) Class B
C) Class C
D) Class D

Correct Answer: C (Class C)

Explanation: Fires involving electrical equipment are classified as Class C fires. A radio transmitter is an electrical device, making any fire within it a Class C fire, requiring extinguishing agents that are non-conductive.

Q2. Containers are manufactured as per standards set by:

A) IMO
B) SOLAS
C) ISO
D) MARPOL

Correct Answer: C (ISO)

Explanation: The International Organization for Standardization (ISO) sets standards for various products, including containers. These standards ensure uniformity, safety, and interoperability in the manufacturing and use of containers globally.

Q3. What, when removed, will assist in the self-extinguishment of a fire?

A) Nitrogen
B) Carbon dioxide
C) Oxygen
D) Argon

Correct Answer: C (Oxygen)

Explanation: Oxygen is a key component of the fire triangle; its removal breaks the combustion process. Therefore, removing oxygen (Option C) will assist in the self-extinguishment of a fire by starving it of this essential element.

Q4. Classification of fire bulkheads inside accommodation spaces is based on fire resisting capabilities. They are classified as:

A) Class A, B, C
B) Class X, Y, Z
C) Class 1, 2, 3
D) Class H, J, K

Correct Answer: A (Class A, B, C)

Explanation: SOLAS classifies fire integrity of bulkheads and decks into Class A, B, and C, based on their fire-resisting properties and the duration they can withstand a standard fire test. Class A offers the highest integrity, followed by B and C.

Q5. Good housekeeping on a vessel prevents fire by:

A) Improving the speed of bilge pumps
B) Eliminating potential fuel sources
C) Cooling electrical control systems
D) Increasing structural fire boundaries

Correct Answer: B (Eliminating potential fuel sources)

Explanation: Good housekeeping is fundamental to fire prevention as it involves keeping spaces clean and orderly, thereby eliminating or minimizing potential fuel sources such as oily rags, waste, and accumulated debris.

Q6. "Below lower flammable limit" means:

A) The vapor mixture is highly volatile and ready to explode
B) It is a mixture of hydrocarbon gases which is not sufficient to initiate a fire (too lean)
C) The atmosphere is completely saturated with fuel vapors (too rich)
D) There is insufficient structural oxygen available for combustion

Correct Answer: B (It is a mixture of hydrocarbon gases which is not sufficient to initiate a fire (too lean))

Explanation: "Below lower flammable limit" means the concentration of flammable vapor in the air is too low to sustain combustion. The mixture is too lean, and an ignition source will not cause it to burn or explode.

Q7. The oxidation process creates:

A) Cooling vapors
B) Heavy scale deposits
C) Heat
D) Free radical inhibitors

Correct Answer: C (Heat)

Explanation: Oxidation is a chemical process that involves the loss of electrons and typically the gain of oxygen. In the context of fire, this process releases energy, primarily in the form of heat.

Q8. Class A divisions are designed to prevent the passage of flame and smoke to the end of a one-hour standard fire test.

A) True
B) False

Correct Answer: A (True)

Explanation: This statement is true. SOLAS regulations require Class A divisions to be capable of preventing the passage of flame and limiting the temperature rise for a standard one-hour fire test, ensuring structural integrity and compartmentation.

Q9. An important tool for fire prevention is:

A) Periodic paint applications
B) Cleaning and good housekeeping
C) Keeping internal ventilation doors wide open
D) Increasing high-expansion foam reserves

Correct Answer: B (Cleaning and good housekeeping)

Explanation: Effective cleaning and good housekeeping practices are essential for fire prevention. They reduce the amount of combustible material available to fuel a fire and help identify potential hazards early.

Q10. The tendency or ability of a liquid to vaporize is called:

A) Viscosity
B) Volatility
C) Density
D) Capillarity

Correct Answer: B (Volatility)

Explanation: Volatility refers to the tendency of a substance, particularly a liquid, to vaporize or form gases at a given temperature. Liquids with high volatility evaporate readily, posing a greater fire risk.

Q11. A person may operate an air compressor in which of the following areas on board a tank barge?

A) Inside a cargo tank hatch
B) Pumproom space
C) Generator room
D) Open deck near vapor manifold lines

Correct Answer: C (Generator room)

Explanation: Operating electrical equipment like an air compressor should ideally be done in a safe area free from flammable vapors. A generator room (Option C) is typically well-ventilated and designed for such equipment, unlike cargo tanks or pumprooms which may contain flammable gases.

Q12. The most efficient way of transferring heat in liquids and gases is called:

A) Conduction
B) Convection
C) Radiation
D) Insulation

Correct Answer: B (Convection)

Explanation: Convection is the transfer of heat through the movement of fluids (liquids or gases). As the fluid heats up, it becomes less dense and rises, while cooler, denser fluid sinks, creating a continuous circulation pattern.

Q13. Fire and lifeboat stations are required to be listed on the:

A) Logbook register
B) Master list (station bill)
C) Cargo manifest layout
D) Crew contract binder

Correct Answer: B (Master list (station bill))

Explanation: The Master List, also known as the Station Bill, details essential information for the crew, including muster stations, fire stations, and lifeboat assignments, as required by maritime regulations like SOLAS.

Q14. Radiation is a process of heat transfer where no mass is exchanged and no medium is required.

A) True
B) False

Correct Answer: A (True)

Explanation: This statement is true. Radiation is a mode of heat transfer that does not require a medium and can travel through a vacuum, such as heat from the sun reaching the Earth or heat radiating from a fire.

Q15. You can find the location of your abandon ship post by checking the:

A) Deck logbook entries
B) Bridge passage plan
C) Master list
D) Safety equipment certificate booklet

Correct Answer: C (Master list)

Explanation: The Master List or Station Bill, which is posted on board, indicates the designated abandon ship post for each crew member. This ensures everyone knows their assigned location during an emergency.

Q16. In order to achieve the fire safety objective, which of the following functional requirements are embodied in SOLAS regulations?

A) Division of the ship into main horizontal/vertical zones by thermal and structural boundaries
B) Separation of accommodation spaces from the remainder of the ship by thermal and structural boundaries
C) Protective means of escape and access for firefighting
D) All of the above

Correct Answer: D (All of the above)

Explanation: SOLAS Chapter II-2 outlines comprehensive fire safety measures. These include dividing the ship into zones, separating accommodation from machinery spaces, and ensuring adequate means of escape and fire-fighting access, encompassing all listed requirements.

Q17. The vessel's Fire Control plan is laid out on which of the following types of plan?

A) Midship structural section
B) General arrangement
C) Lines plan drawing
D) Tank capacity schematic

Correct Answer: B (General arrangement)

Explanation: The Fire Control Plan, mandated by SOLAS, is typically a detailed General Arrangement plan showing the location of fire detection systems, fire-fighting equipment, means of escape, and boundaries of fire zones.

Q18. Carbon dioxide is stored under pressure in liquid form inside a fire:

A) Hose connection reel
B) Foam applicator drum
C) Cylinder
D) Storage box enclosure

Correct Answer: C (Cylinder)

Explanation: Carbon dioxide (CO2) for fire suppression is stored under pressure in liquid form within robust cylinders, which are part of the fixed or portable fire extinguishing systems.

Q19. Which statement concerning carbon dioxide is FALSE?

A) It leaves no dynamic residue on machinery components
B) It is safe to use near personnel in a confined space
C) It extinguishes fire by smothering action
D) It can be utilized safely on electrical fires

Correct Answer: B (It is safe to use near personnel in a confined space)

Explanation: While CO2 is effective and safe for electrical fires, it displaces oxygen and can cause asphyxiation. It is NOT safe to use near personnel in a confined space without proper respiratory protection and prior ventilation.

Q20. Which class division is considered capable of preventing the passage of ONLY flame to the end of the first half-hour of a standard fire test?

A) Class A
B) Class B
C) Class C
D) Class D

Correct Answer: B (Class B)

Explanation: Class B divisions are designed according to SOLAS regulations to prevent the passage of flame for the first half-hour of a standard fire test. This provides crucial time for evacuation and initial firefighting efforts.

Q21. A fire hose can be used for purposes other than firefighting service when:

A) Washing down cargo decks at sea
B) Used for drills and testing
C) Supplying cooling fluid to a temporary generator
D) Clearing choked scupper valves

Correct Answer: B (Used for drills and testing)

Explanation: Fire hoses are critical safety equipment and should primarily be used for firefighting. Their use for drills and testing (Option B) is permissible and necessary for ensuring their readiness, but other uses can compromise their integrity.

Q22. How many CO2 fire extinguishers ought to be kept in accommodation areas?

A) One per room corridor
B) No extinguishers
C) Two near the galley spaces
D) Four on each deck landing

Correct Answer: B (No extinguishers)

Explanation: CO2 extinguishers are generally not recommended for accommodation areas due to the risk of asphyxiation in confined spaces. Water-based or foam extinguishers are more commonly found in these areas, with specific types and numbers dictated by regulations.

Q23. Portable foam fire extinguishers are designed for use on which classes of fires?

A) Class A and B fires
B) Class B and C fires
C) Class C and D fires
D) Class A and D fires

Correct Answer: A (Class A and B fires)

Explanation: Portable foam fire extinguishers are primarily designed for Class A fires (ordinary combustibles) and Class B fires (flammable liquids). They work by creating a foam blanket that smothers the fire and cools the fuel.

Q24. The removal of fuel in the firefighting process is known as:

A) Smothering
B) Cooling
C) Starvation
D) Inhibition

Correct Answer: C (Starvation)

Explanation: Starvation is the fire extinguishment technique of removing the fuel source. By cutting off the supply of fuel to the fire, the combustion process ceases.

Q25. Which of the following classes of fire would probably occur in the engine room bilges?

A) Class A
B) Class B
C) Class C
D) Class D

Correct Answer: B (Class B)

Explanation: Engine room bilges often contain flammable liquids like lubricating oil, hydraulic oil, or fuel oil. A fire involving these substances is classified as a Class B fire.

Q26. Why is a CO_2 type of fire extinguisher preferred for electrical installations?

A) It provides extensive long-term cooling
B) It expands into a sticky visual foam barrier
C) It leaves no residue and does not damage any equipment
D) It can be safely deployed without isolating power circuits

Correct Answer: C (It leaves no residue and does not damage any equipment)

Explanation: CO_2 is preferred for electrical installations because it is a non-conductive gas that extinguishes fire by displacing oxygen and cooling, and importantly, it leaves no residue that could damage sensitive electronic components.

Q27. Which of the listed extinguishing agents is NOT suitable for fighting a liquid paint fire?

A) Dry chemical powder
B) Carbon dioxide gas
C) Water
D) Mechanical foam spray

Correct Answer: C (Water)

Explanation: Water is generally unsuitable for liquid paint fires (Class B) as it can spread the burning liquid due to its immiscibility and lower density, potentially intensifying the fire. Foam, dry chemical, and CO_2 are more appropriate agents.

Q28. As per NFPA (National Fire Protection Association), a fire in cooking appliances involving vegetable/animal oils and fats is classified as:

A) Class B
B) Class D
C) Class C
D) Class K

Correct Answer: D (Class K)

Explanation: Fires involving cooking appliances and the specific oils and fats used in them are classified as Class K fires under the NFPA system, distinct from Class B which covers general flammable liquids.

Q29. Foam is effective in combating which classes of fire?

A) Class A and B
B) Class B and C
C) Class C and D
D) Class A and D

Correct Answer: A (Class A and B)

Explanation: Foam extinguishers work by creating a smothering blanket that separates the fuel from oxygen, effectively suppressing Class A (solid combustibles) and Class B (flammable liquids) fires. It is generally not suitable for Class C (electrical) due to conductivity or Class D (metal) fires.

Q30. Designated areas for smoking are specified in the:

A) International code manual
B) Company manual
C) Port state entry document
D) Crew mess instruction card

Correct Answer: B (Company manual)

Explanation: Designated smoking areas are safety-critical zones established by the company's Safety Management System (SMS) to prevent fire hazards. These are clearly defined in the company manual and shipboard standing orders to ensure compliance with international safety standards.

Practice Set 5

Q1. The spreading of fire as a result of heat being carried through a vessel's ventilation system is an example of heat transfer by:

A) Conduction
B) Convection
C) Radiation
D) Absorption

Correct Answer: B (Convection)

Explanation: Convection is the transfer of heat through the movement of fluids (liquids or gases). In a vessel's ventilation system, hot air and gases rise and circulate, carrying heat to other parts of the ship and potentially spreading fire.

Q2. On cargo and miscellaneous vessels, what is NOT a required part of the fireman's outfit?

A) Protective clothing
B) Rigid boots and gloves
C) Combustible gas indicator
D) Flame safety lamp or breathing apparatus

Correct Answer: C (Combustible gas indicator)

Explanation: A combustible gas indicator is used to detect flammable gases, which is not a standard component of the basic fireman's outfit required by SOLAS. The outfit focuses on protection from heat, flames, and providing breathing apparatus for toxic atmospheres.

Q3. A fire hose kept in its box should have a wrench and an:

A) All-purpose nozzle
B) High expansion foam generator tube
C) International Shore standard coupling flange
D) High-pressure mechanical lance tool

Correct Answer: A (All-purpose nozzle)

Explanation: An all-purpose nozzle allows for adjustment of the water stream from a spray to a solid jet, which is essential for effective firefighting. While other items might be associated with firefighting equipment, the nozzle is a primary component of a hose box.

Q4. Actuating the fixed CO2 system should cause the automatic shutdown of the:

A) Main engine propulsion controls
B) Emergency bilge pump circuits
C) Supply and exhaust ventilation
D) Main switchboard power grid distribution

Correct Answer: C (Supply and exhaust ventilation)

Explanation: Activating a fixed CO2 system is designed to extinguish fires by displacing oxygen. To prevent re-ignition and potential damage, it automatically shuts down ventilation fans that could supply air to the fire or spread the gas. Engine propulsion and power distribution are typically handled manually or through other safety interlocks.

Q5. During a fire onboard, communication is to be:

A) Complex, thoroughly detailed, and extensive
B) Concise, precise, and short
C) Relayed solely via messengers
D) Postponed until structural assessments end

Correct Answer: B (Concise, precise, and short)

Explanation: During a fire, clear and rapid communication is paramount for effective response and safety. Messages must be concise and precise to avoid confusion, ensuring that critical information is understood quickly by all involved parties.

Q6. A solid stream of water is produced by an all-purpose firefighting nozzle when the handle is:

A) Pushed all the way forward
B) Positioned halfway into its casing
C) Pulled all the way back
D) Twisted ninety degrees counterclockwise

Correct Answer: C (Pulled all the way back)

Explanation: On most firefighting nozzles, pulling the handle back is the action that adjusts the nozzle to produce a solid stream of *water. Pushing it forward typically creates a spray pattern, which is also useful for cooling but not for a solid stream.*

Q7. Immediately after extinguishing a fire with CO2, it is advisable to:

A) Restore structural compartment ventilation fully
B) Open up all access ports to inspect inside details
C) Stand by with water or other agents
D) Drain internal oil recovery sumps directly

Correct Answer: C (Stand by with water or other agents)

Explanation: CO2 displaces oxygen and can cause asphyxiation. Immediately after use, there is a risk of re-flash if the heat source remains, or re-ignition due to the presence of unburnt fuel. Therefore, standing by with backup extinguishing agents is crucial.

Q8. Ships carrying more than 36 passengers should maintain a:

A) Continuous automated water system monitor
B) Fire patrol system
C) Double bulkhead smoke containment grid
D) Dedicated fire drill barge trailer connection

Correct Answer: B (Fire patrol system)

Explanation: SOLAS regulations require passenger ships carrying more than 36 passengers to maintain a fire patrol system. This involves regular patrols by crew members to detect and report fire hazards, ensuring prompt response.

Q9. During cargo operations, a deck fire has occurred due to a leaking cargo line. You should first:

A) Rig a mechanical foam generator station
B) Sound the international collision alarm whistle
C) Stop the transfer of cargo
D) Douse structural pipelines with sea spray

Correct Answer: C (Stop the transfer of cargo)

Explanation: During cargo operations, the immediate priority upon discovering a fire related to cargo transfer is to stop the source of the fuel. Stopping the cargo transfer prevents further spillage and fuel supply to the fire.

Q10. Radio station logs involving communication during a disaster shall be kept by the station licensee for at least:

A) One year from entry initiation
B) Three years from the date of the last entry
C) Five years under legal maritime seals
D) Ten years inside safe archive storage boxes

Correct Answer: B (Three years from the date of the last entry)

Explanation: Radio station logs, particularly those pertaining to emergency communications, must be maintained for a specified period to serve as official records for investigations or audits. SOLAS and relevant maritime regulations typically mandate a retention period of three years from the date of the last entry.

Q11. The deep fryer oil overheated because of a:

A) Lack of fresh liquid fat inputs
B) Thermostat malfunctioning
C) Mechanical failure inside the extractor canopy
D) Blockage inside draining pipe lines

Correct Answer: B (Thermostat malfunctioning)

Explanation: Deep fryer fires are often caused by overheating oil, which can result from a malfunctioning thermostat. If the thermostat fails to regulate the temperature, the oil can reach its smoke point and then its flashpoint, leading to ignition.

Q12. The best way to test the Inmarsat-C terminal is to:

A) Place a voice phone link call to shore coordination networks
B) Compose and send a brief message to your own Inmarsat-C terminal
C) Sound the internal automatic distress loop button
D) Contact an adjacent vessel via bridge VHF radio link

Correct Answer: B (Compose and send a brief message to your own Inmarsat-C terminal)

Explanation: The most effective way to test the functionality of an Inmarsat-C terminal, including its ability to transmit and receive messages, is to send a test message to itself or a known contact. This verifies the communication path and system response without relying on external systems.

Q13. The control panel of a fire detection system must have all of the following EXCEPT:

A) Visual indicators showing alarm sectors
B) An audible device to signal alarms or system faults
C) A way to bypass the entire panel if it malfunctions
D) Test switches to check structural loop indicators

Correct Answer: C (A way to bypass the entire panel if it malfunctions)

Explanation: Fire detection system control panels are designed with safety and operational indicators, including ways to silence alarms, reset the system, and identify the zone of activation. Bypassing the entire panel is a safety risk and not a standard feature; systems should remain functional or be manually overridden at zone levels if necessary.

Q14. Tapioca is susceptible to ________ when damp.

A) Chemical crystallization
B) Liquefaction slide effects
C) Spontaneous combustion
D) Electrostatic discharges

Correct Answer: C (Spontaneous combustion)

Explanation: Tapioca, like many organic materials, can undergo self-heating when damp due to microbial activity or chemical reactions. If this heat cannot dissipate, it can lead to spontaneous combustion, igniting the material.

Q15. As a firefighting medium, CO2 can be dangerous under certain conditions as it can cause:

A) Severe toxic vapor contamination inside boiler loops
B) Freeze burns and blistering
C) Spontaneous flashover due to compression waves
D) Corrosion of sensitive mechanical gears

Correct Answer: B (Freeze burns and blistering)

Explanation: CO2 is stored under pressure and expands rapidly when released, causing a significant drop in temperature. This extreme cold can cause freeze burns or blistering on contact with skin, and also reduces visibility by creating fog.

Q16. As per NFPA, a fire in an electrical generator is considered to be a:

A) Class A fire
B) Class B fire
C) Class C fire
D) Class D fire

Correct Answer: C (Class C fire)

Explanation: Fires involving electrical equipment, such as generators, are classified as Class C fires. This classification specifically denotes fires that involve energized electrical equipment where the use of a conductive extinguishing agent could lead to electrocution.

Q17. If passengers are on board when an abandon ship drill is carried out, they should:

A) Remain inside their cabins behind closed safety doors
B) Gather exclusively at public dining lounges
C) Take part
D) Be evacuated on an auxiliary service tender craft

Correct Answer: C (Take part)

Explanation: During abandon ship drills, all persons on board, including passengers, are expected to participate. This ensures they are familiar with procedures, muster stations, and the use of safety equipment, aligning with SOLAS requirements.

Q18. On the vessel's Fire Control plan, all parts of a fixed fire suppression system are listed EXCEPT:

A) Location of piping distribution loops
B) Total weight and type of chemical storage canisters
C) Instructions for activation of the system
D) Placement of discharge nozzles across compartments

Correct Answer: C (Instructions for activation of the system)

Explanation: Fire control plans, as mandated by SOLAS, detail the layout, locations of fire detection and suppression systems, and emergency equipment. While they show system components and operational instructions, specific activation procedures are usually found in operational manuals rather than on the plans themselves.

Q19. Through which of the listed processes is sufficient heat produced to cause spontaneous ignition?

A) Heat of condensation
B) Heat of vaporization
C) Heat of oxidation
D) Heat of radiation

Correct Answer: C (Heat of oxidation)

Explanation: Spontaneous ignition occurs when combustible materials generate enough heat internally to reach their ignition temperature without an external ignition source. This process is primarily driven by the heat of oxidation, where slow chemical reactions release heat over time.

Q20. Fusible link fire dampers are operated by:

A) Electronic loop control commands from the safety desk
B) The heat of the fire melting the link
C) Hydrostatic water pressure changes from water spray systems
D) Structural balance wires linked with the bridge desk

Correct Answer: B (The heat of the fire melting the link)

Explanation: Fusible link fire dampers are designed to close automatically when the temperature reaches a certain point. The heat *from a fire melts a solder or alloy link, allowing the damper to fall shut and restrict airflow, thus preventing fire and smoke spread.*

Q21. You have just extinguished an oil fire on the floor plates of the engine room with a 15-pound CO2 extinguisher. Which of the listed dangers should you now be preparing to handle?

A) Complete systemic failure of main engine gear oil feeds
B) Toxic gas ignition inside air trunk lines
C) Reflashing of the fire
D) Frostbite cracks inside fuel line connection joints

Correct Answer: C (Reflashing of the fire)

Explanation: CO2 extinguishes fire by displacing oxygen but does not cool the fuel significantly. Therefore, any remaining hot spots or fuel near the ignition temperature can easily re-flash once the CO2 dissipates and oxygen returns.

Q22. The height of a VHF radio antenna is more important than the power output wattage of the radio because:

A) Skywave atmospheric returns are highly unstable at sea
B) VHF communications are basically line of sight
C) Higher placement reduces tracking ground interference loops
D) Internal wire resistances limit low-level signal paths

Correct Answer: B (VHF communications are basically line of sight)

Explanation: VHF radio waves travel in straight lines and are limited by the curvature of the Earth; thus, they are fundamentally line-of-sight communications. Antenna height is crucial for extending this line of sight and improving the range of communication.

Q23. Which extinguishing agent is most likely to allow reflash as a result of not cooling the fuel below its ignition temperature?

A) Aqueous film-forming foam (AFFF)
B) CO2
C) Water fog stream
D) Dry chemical powder mix

Correct Answer: B (CO2)

Explanation: CO2 is a gaseous extinguishing agent that works by displacing oxygen and cooling. However, it does not effectively cool the burning fuel below its ignition temperature, making reflash a significant risk once the CO2 disperses.

Q24. A vessel's Fire Control plan shall have:

A) Blueprint files locked exclusively in the captain's office
B) A duplicate set of plans permanently stored outside the deck house in a prominent weatherproof enclosure
C) Microfiche summary records mounted inside all lifeboats
D) Plastic layout cards stapled on every accommodation deck door

Correct Answer: B (A duplicate set of plans permanently stored outside the deck house in a prominent weatherproof

Explanation: SOLAS regulations require fire control plans to be readily accessible to crew and emergency responders. A duplicate set stored in a prominent, weatherproof enclosure outside the navigation bridge ensures vital information is available even if the bridge is inaccessible.

Q25. In the regulations for cargo vessels, which statement is true concerning a fireman's outfit?

A) Each outfit must contain an electric welding cutting tool
B) Each fireman's outfit shall contain a flame safety lamp of an approved type
C) Air tanks must carry five hours of continuous storage reserves
D) Two outfits are needed for each active deck crewman

Correct Answer: B (Each fireman's outfit shall contain a flame safety lamp of an approved type)

Explanation: According to SOLAS regulations, each fireman's outfit must include a flame safety lamp or an approved self-contained breathing apparatus. This is to ensure the safety of the firefighter in potentially hazardous and oxygen-deficient atmospheres.

Q26. Fire alarm system thermostats are actuated by:

A) Air pressure changes inside trunk compartments
B) The difference in thermal expansion of two dissimilar metals
C) Light scattering effects across laser sensors
D) Moisture tracking limits inside structural wiring panels

Correct Answer: B (The difference in thermal expansion of two dissimilar metals)

Explanation: Thermostats used in fire alarm systems, particularly bimetallic types, operate based on the principle that two different metals expand at different rates when heated. This differential expansion causes the strip to bend, closing an electrical contact and triggering an alarm.

Q27. For the safety of the personnel working with fire hoses, the fire pumps are fitted with a:

A) Remote automated vacuum seal breaker
B) Mechanical safety clutch linkage on driving engines
C) Pressure gauge and relief valve on the discharge side
D) Flexible expansion joint directly on the water suction intake

Correct Answer: C (Pressure gauge and relief valve on the discharge side)

Explanation: Fire pumps are equipped with pressure gauges to monitor discharge pressure and relief valves to prevent over-pressurization, which could damage the pump, hoses, or injure personnel. This is a critical safety feature ensuring stable operation.

Q28. Dry chemical extinguishers extinguish Class B fires to the greatest extent by:

A) Cooling fuel temperatures down below flashpoint margins
B) Displacing essential atmospheric nitrogen compounds
C) Breaking the chain reaction
D) Freezing burning vapor pools immediately

Correct Answer: C (Breaking the chain reaction)

Explanation: Dry chemical powders interrupt the chemical chain reaction of a fire, effectively stopping the combustion process. While they have some cooling effect, their primary extinguishing mechanism for Class B and C fires is the breaking of the chain reaction.

Q29. What shall be conducted during a fire and boat drill?

A) Cargo tank hatches must be opened for visual gas checking
B) All watertight doors in the vicinity of the drill shall be operated
C) Emergency bilge suctions must be run on full backflow parameters
D) Lifeboats must be launched and run at service speeds for an hour

Correct Answer: B (All watertight doors in the vicinity of the drill shall be operated)

Explanation: During a combined fire and boat drill, operational checks of essential safety equipment are conducted. Operating *watertight doors in the vicinity of the drill ensures their functionality and familiarizes the crew with their operation under simulated emergency conditions.*

Q30. When approaching a fire from windward, you should shield firefighters from the fire by using an applicator and:

A) High velocity solid water jet
B) High expansion foam blanket throw
C) Low velocity fog
D) CO2 continuous vapor jet

Correct Answer: C (Low velocity fog)

Explanation: When fighting a fire from windward, using a low-velocity fog pattern from a nozzle applicator helps shield firefighters from radiant heat and smoke. The water droplets absorb heat and create a screen, allowing safer approach and application of the extinguishing agent.

Practice Set 6

Q1. The extinguishing agent most likely to allow re-ignition of a fire is:

A) Mechanical foam spray
B) Carbon dioxide
C) Water fog
D) Dry chemical powder

Correct Answer: B (Carbon dioxide)

Explanation: Carbon dioxide (CO2) extinguishes fire by displacing oxygen. However, it has minimal cooling effect, meaning that if the fuel remains hot enough, it can re-ignite once the CO2 dissipates and oxygen is reintroduced.

Q2. The risk of fire and explosion is ________ in tankers.

A) Negligible under standard cruising speeds
B) Very high
C) Identical to generic dry cargo vessels
D) Controlled solely by hull thickness layout

Correct Answer: B (Very high)

Explanation: Tankers, especially those carrying petroleum products, handle large quantities of flammable liquids. These cargoes create a very high risk of fire and explosion, necessitating stringent safety measures and procedures as per MARPOL and other regulations.

Q3. Why should foam be banked off a bulkhead when extinguishing an oil fire?

A) To speed up chemical stabilization loops of active foam bubbles
B) To prevent agitation of the oil and spreading the fire
C) To maximize the coverage range of standard delivery equipment
D) To avoid cooling structural steel components unevenly

Correct Answer: B (To prevent agitation of the oil and spreading the fire)

Explanation: When applying foam to an oil fire, banking it off a bulkhead prevents the direct force of the foam stream from agitating the oil surface. Agitation can spread the burning oil and potentially worsen the fire, hindering extinguishment.

Q4. A large oil fire on the deck of a ship can be fought most effectively with:

A) Solid streams of sea water
B) High pressure carbon dioxide gas grids
C) Foam
D) Portable dry powder spray canisters

Correct Answer: C (Foam)

Explanation: Foam is the most effective agent for fighting large oil fires on decks because it spreads across the surface, cools the fuel, and smothers the fire by forming a blanket that separates the fuel from the oxygen.

Q5. The minimum flashpoint of oil fuel carried on board a ship is:

A) 40 degrees Celsius
B) 50 degrees Celsius
C) 60 degrees Celsius
D) 70 degrees Celsius

Correct Answer: C (60 degrees Celsius)

Explanation: According to the International Maritime Dangerous Goods (IMDG) Code and ISM Code, the minimum flashpoint for fuel oil permitted for carriage on board ships is generally 60 degrees Celsius (closed cup).

Q6. You are operating a fire hose with an applicator attached. If you put the handle of the nozzle in the vertical position, you will use:

A) High velocity solid jet stream
B) Shutoff position with zero output flow
C) Low velocity fog
D) Foam solution delivery parameter

Correct Answer: C (Low velocity fog)

Explanation: With an applicator attached to a fire hose nozzle, placing the handle in the vertical position typically adjusts the nozzle to produce a low-velocity fog pattern. This is effective for cooling and creating a protective screen.

Q7. To safely enter a compartment where CO2 has been released from a fixed extinguishing system, you should:

A) Wear a generic particulate dust filter mask
B) Run a hand-held exhaust blower for five minutes first
C) Wear a self-contained breathing apparatus (SCBA)
D) Wait ten minutes for gas settlement before walking in

Correct Answer: C (Wear a self-contained breathing apparatus (SCBA))

Explanation: Carbon dioxide (CO2) displaces oxygen and can create a hazardous atmosphere for breathing. Entering a compartment where CO2 has been released requires a self-contained breathing apparatus (SCBA) to ensure a safe supply of breathable air.

Q8. The international shore connection:

A) Links ship navigation logs with port administration grids
B) Allows hookup of firefighting water from shore facilities
C) Supplies high voltage electrical power to propulsion links
D) Drains oily bilge slops safely to dockside collection tanks

Correct Answer: B (Allows hookup of firefighting water from shore facilities)

Explanation: The international shore connection is a standardized coupling that allows a ship's fire main system to be connected to shore-based firefighting water supplies. This provides an external source of water in emergencies.

Q9. A fire has broken out on the stern of your vessel. You should maneuver your vessel so the wind:

A) Comes directly over the stern layout beam
B) Blows at right angles to cross-vent the hull decks
C) Comes over the bow
D) Changes into circular thermal swirling patterns

Correct Answer: C (Comes over the bow)

Explanation: When fighting a fire on a vessel, positioning the ship so the wind blows the fire away from you is crucial. Approaching the fire with the wind coming over the bow allows the wind to help push the flames and heat away from the firefighting crew.

Q10. An extinguishing agent which effectively cools, dilutes combustible vapors, removes oxygen, and provides a heat and smoke screen is:

A) Halon gas
B) Dry chemical powder formulation
C) Water fog
D) Aqueous film-forming foam solution

Correct Answer: C (Water fog)

Explanation: Water fog, when applied correctly, has multiple extinguishing properties. It cools the fuel below its ignition temperature, dilutes combustible vapors, displaces oxygen, and provides a screen against radiant heat and smoke.

Q11. The expansion of LFL is:

A) Liquid Fuel Line
B) Low Flashpoint Liquid
C) Lower Flammable Limit
D) Light Flame Location

Correct Answer: C (Lower Flammable Limit)

Explanation: LFL stands for Lower Flammable Limit, which is the minimum concentration of a flammable gas or vapor in air that will burn or explode when an ignition source is present. This concept is critical for assessing fire and explosion hazards.

Q12. Which statement describes the primary process by which fires are extinguished by dry chemical?

A) The dry chemical creates a dense visual foam overlay block
B) The dry chemical powder attacks the fuel and oxygen chain reaction
C) The active agent drops fuel core parameters past absolute freezing margins
D) It relies on heavy moisture transformation into structural vapor screens

Correct Answer: B (The dry chemical powder attacks the fuel and oxygen chain reaction)

Explanation: Dry chemical powders primarily extinguish fires by interrupting the chemical chain reaction of combustion. They also have a slight cooling effect and can form a barrier between the fuel and oxygen.

Q13. What is the most important consideration when determining to fight an electrical fire?

A) The replacement costs of burnt circuitry components
B) The clean-up time required for active extinguishing residue
C) Danger of shock to personnel
D) Shutting down auxiliary lighting networks across accommodation spaces

Correct Answer: C (Danger of shock to personnel)

Explanation: When fighting electrical fires, the primary concern is the risk of electric shock to personnel. This risk necessitates the use of appropriate extinguishing agents that do not conduct electricity.

Q14. If you see an oil leakage, you clean it immediately. This is an example of:

A) Preventive machinery tuning parameters
B) ISO structural standard declaration rules
C) Good housekeeping
D) Deck log monitoring instructions

Correct Answer: C (Good housekeeping)

Explanation: Promptly cleaning up oil leakages is an essential practice known as good housekeeping. Maintaining a clean and orderly environment prevents slip hazards and potential fire risks.

Q15. When dry chemical extinguishers are used to put out Class B fires, there is a danger of reflash because dry chemical:

A) Reacts aggressively with marine steel materials
B) Does little or no cooling
C) Displaces air structures for only a few seconds
D) Vaporizes into flammable gas compounds under heat exposure

Correct Answer: B (Does little or no cooling)

Explanation: Dry chemical agents are effective at suppressing Class B fires by interrupting the chemical reaction, but they offer minimal cooling. This lack of cooling capability means that the fuel can still vaporize and re-ignite if the heat source persists.

Q16. Which of the following fire extinguishing agents has the greatest capacity for heat absorption?

A) Carbon dioxide gas jet
B) Dry chemical powder spray
C) Water fog
D) Mechanical foam blanket expansion

Correct Answer: C (Water fog)

Explanation: Water fog, or spray, has the greatest capacity for heat absorption among the listed agents due to water's high latent heat of vaporization. This rapid cooling is crucial for extinguishing fires.

Q17. The expansion of IBC code is:

A) International Bulk Cargo code
B) International Bulk Chemical code
C) Intermaritime Boiler Construction code
D) International Bilge Control code

Correct Answer: B (International Bulk Chemical code)

Explanation: The IBC Code, or International Bulk Chemical Code, provides regulations for the carriage of dangerous chemicals in bulk by sea. It is part of the IMO's framework for maritime safety and environmental protection.

Q18. As compared to carbon dioxide, dry chemical has which advantage?

A) It leaves no visible residue cleanup footprint on components
B) It offers significant localized thermal cooling effects
C) Greater range
D) It is non-toxic inside confined spaces

Correct Answer: C (Greater range)

Explanation: Dry chemical extinguishers have a longer discharge range compared to carbon dioxide extinguishers. This allows firefighters to tackle fires from a safer distance.

Q19. Which of the listed firefighting agents and associated applications provide the largest shielding effect for the firefighter?

A) Straight high-velocity water jet
B) High-expansion chemical foam spray
C) Low velocity water fog
D) Continuous carbon dioxide stream jet

Correct Answer: C (Low velocity water fog)

Explanation: Low velocity water fog provides a significant shielding effect for firefighters by absorbing heat and producing steam, which displaces oxygen and limits flame spread. This creates a safer working environment.

Q20. An example of a type of smoke detector is the:

A) Thermal differential bimetallic link switch
B) Ionization type
C) Hydrostatic expansion capsule sensor
D) Visual infrared flame scanner screen

Correct Answer: B (Ionization type)

Explanation: The ionization type smoke detector is a common fire detection device that works by monitoring changes in electrical conductivity within an ionization chamber.

Q21. Which firefighting agent is most effective in removing heat?

A) Carbon dioxide vapor stream
B) Dry chemical powder blanket throw
C) Water spray
D) Mechanical structural foam solution

Correct Answer: C (Water spray)

Explanation: Water spray, including water fog, is the most effective firefighting agent for removing heat from a fire. Its high latent heat of vaporization allows it to absorb a significant amount of thermal energy as it turns into steam.

Q22. When water fog is used as an extinguishing agent, the fire is extinguished principally by the:

A) Displacement of active fuel structures safely
B) Absorption of heat
C) Exclusion of atmospheric nitrogen elements
D) Chemical breakdown of active free radicals

Correct Answer: B (Absorption of heat)

Explanation: When water fog is applied to a fire, its primary extinguishing mechanism is the rapid absorption of heat. As the water vaporizes into steam, it cools the fuel below its ignition temperature.

Q23. While fighting a coal fire, forced ventilation can be used to bring down the temperature.

A) True
B) False

Correct Answer: B (False)

Explanation: Forced ventilation should not be used when fighting a coal fire, as it can supply more oxygen and intensify the combustion process. Smothering the fire is generally the preferred method.

Q24. Fire extinguishers of sizes 3, 4, and 5 are designated as:

A) Hand portable
B) Semi-portable
C) Fixed system bank units
D) Auxiliary utility modules

Correct Answer: B (Semi-portable)

Explanation: Fire extinguishers with sizes 3, 4, and 5 (typically kilograms or pounds) are classified as semi-portable. They are too heavy to be easily carried by one person but can be moved with effort.

Q25. Dangerous goods are classified into how many groups?

A) Three groups
B) Five groups
C) Nine groups
D) Twelve groups

Correct Answer: C (Nine groups)

Explanation: Dangerous goods, as defined by international regulations like the IMDG Code, are classified into nine hazard classes based on their properties and the risks they pose.

Q26. Fire safety is achieved by using fire retardant materials.

A) True
B) False

Correct Answer: A (True)

Explanation: Fire safety is indeed achieved by using fire retardant materials, which are designed to resist ignition and slow down the spread of fire. This is a key principle in ship construction and fire prevention.

Q27. The discharge from a carbon dioxide fire extinguisher should be directed:

A) At the top perimeter layout of smoke clouds
B) At the base of the flames
C) Directly into adjacent ventilation return trunks
D) In wide looping circles across structural partition bulkheads

Correct Answer: B (At the base of the flames)

Explanation: The discharge from a carbon dioxide fire extinguisher should always be directed at the base of the flames. This is where the fuel is being vaporized and combustion is occurring, allowing CO2 to displace oxygen and cool the fuel.

Q28. Which statement is true concerning the use of dry chemical extinguishers?

A) You should direct the spray at the top edges of structural smoke screens
B) You should direct the spray at the base of the fire
C) They are deployed safely on metal fires like sodium or magnesium
D) They provide deep cooling to prevent any risk of re-ignition

Correct Answer: B (You should direct the spray at the base of the fire)

Explanation: When using dry chemical extinguishers, the spray should always be directed at the base of the fire. This action smothers the fire by interrupting the chemical chain reaction and forming a barrier.

Q29. Alcohol-resistant foam is also known as all-purpose foam.

A) True
B) False

Correct Answer: A (True)

Explanation: Alcohol-resistant foam (AR-AFFF) is a type of foam designed to be effective on fires involving polar solvents like alcohols, as well as hydrocarbon fuels. It is often referred to as all-purpose foam due to its versatility.

Q30. What is the minimum number of portable fire extinguishers that must be carried on ships of 1000 GRT and above?

A) Two
B) Three
C) Five
D) Eight

Correct Answer: C (Five)

Explanation: According to SOLAS regulations, ships of 1000 GRT and above must carry a minimum of five portable fire extinguishers, ensuring adequate fire-fighting capability across the vessel.